uality Knowledge as a Driver of Computing Compe-
: An International Project-Based Learning Project

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This paper presents key recommendations developed based on the analysis of an in-
l didactic project conducted jointly by two universities. The initiative aimed to enhance
competencies in data processing systems, software engineering, and the applications of
science in environmental engineering. Utilizing re-al-world air quality data and modern
ies (Node-RED, RabbitMQ, Docker), the project provided students with hands-on
e in near-production environments.

sis of the course outcomes and performance led to the formulation of tar-geted
ndations across three core pillars:

Engineering: Emphasizes deeper integration of the "learning by doing" approach alongside
gineering practices (documentation, automated testing, code quality control) to effectively
pid prototyping with professional industry standards.

ental Application Design: Highlights the necessity of embedding the environmental
the foundation of the design process, recommending the development of user-engaging
gamification, mobile sensor data) to foster citizen science.

versity Cooperation: Advocates for a hybrid teaching model to maximize interaction,
the standardization of evaluation criteria and collaborative tools (e.g., Git) to better
udents for a global professional environment.

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delines offer a strategic framework for optimizing future educational initiatives that bridge actice, and international academic synergy.

s: Air quality, computing competence, system development, Software Engineering, ental Application Design.

roduction

a of progressive climate change and its compounding negative impacts, life in glomerations is becoming increasingly challenging. Urban populations, alongside a and fauna, are highly vulnerable to the amplified frequency of extreme weather nd severe thermal stress. Anthropogenic climate models predict that the global maximum temperature in European, South American, and African cities will by 2 to 8°C over the coming decades [Cl26]. This systemic temperature increase, d with shifting precipitation regimes, triggers a dangerous synergy with localized tion, drastically reducing the environmental carrying capacity of modern cities

ld Health Organization (WHO) classifies ambient air pollution as one of the most environmental hazards to human health. Prolonged exposure to elevated ations of atmospheric pollutants induces severe systemic health failures, nantly manifesting as chronic respiratory diseases, acute asthma exacerbations, emic strokes. Highlighting the scale of this crisis, the European Environment EEA) attributes substantial premature mortality rates directly to poor air quality, nual losses of 253,000 lives due to fine particulate matter ($PM_{2.5}$), 52,000 due to dioxide (NO_2), and 22,000 due to ground-level ozone (O_3) (klimaoslo, 2025).

s is heavily compounded by rapid global demographic shifts. Urbanization acts or driver of local microclimatic modification, often intensifying the Urban Heat HI) effect. Szałata et al. (2024) identify expanding urban structures as complex, ated sources of diverse pollutants, ranging from vehicular exhaust fumes and l emissions to heavy construction dust.

ng these interconnected challenges requires a fundamental realignment of l policies toward sustainable development goals [Gi22]. Derlukiewicz et al. gue that the most effective response to these urban vulnerabilities is the systemic ntation of low-carbon solutions and the structural transition toward "low-carbon meworks. Crucially, as Fu et al. (2010) emphasize[FLW10], low-carbon urban is a multi-objective optimization process. It does not merely seek the reduction house gas emissions, but simultaneously aims to secure decoupling-based growth and elevate the socioeconomic quality of life for citizens. According to prary frameworks, low-carbon initiatives represent an intersection of advanced gies, mitigation strategies, and green practices designed to capture or reduce eric carbon dioxide (CO_2) while fostering long-term urban resilience aha, 2025). Consequently, modern municipalities are increasingly forced to re- their infrastructure—ranging from energy grids to real-time environmental ng networks—to mitigate emissions and safeguard public health.

vith knowledge of air quality and its impact on health, the authors of this on attempted to develop an application solution aimed at implementing an app d enable navigation around the Campus and its surrounding areas in Gdańsk and in areas with the best air quality. The aim of the work was to monitor the most ycling routes and implement them into the operating system.

ors of the publication, together with students, identified the most frequently used ads to the Campus in order to verify them in terms of air quality.

ly developed application for air quality assessment, designed to integrate with available applications, determines optimal access routes based on the nental parameter of air quality. According to the authors, who represent an plinary perspective, this is a novelty that can be confidently implemented into available solutions in the future, constituting a new area of research activity.

lication presents key project assumptions, highlighting the potential of AI tools onmental protection and building useful applications. It also outlines the research logy and expands the chapter on Integration of Web Engineering and nental Education. Importantly, it presents recommendations for implementing the d solutions and key conclusions drawn from the international project.

udent education model

ng internationally coordinated, interdisciplinary student projects bridging , server infrastructure, and artificial intelligence (AI) requires a multifaceted cal approach. Establishing this framework relies on literature covering modern of Things (IoT) pedagogy, distributed project management, AI-assisted coding, hine learning (ML) for environmental data analysis [GA23], [Ka23].

gogy has shifted from lectures to active, Project-Based Learning (PBL) and Maker education. Integrating iterative engineering design—proposing questions, implementing, and evaluating—bridges theoretical knowledge with physical Ch20]. Building tangible environmental monitors connects students directly with lection, making abstract computational concepts accessible and boosting ent [SÇS25].

g complex, multi-university projects requires robust technological and team nent frameworks. Distributed IoT systems provide real-time insights into bottlenecks and student performance, allowing managers to adapt teaching logies and align distributed teams [GA23].

eously, Large Language Models (LLMs) and AI coding assistants (e.g., GitHub ChatGPT) have transformed software engineering. In education, AI tools enhance ming efficiency, though students often struggle with over-reliance and evaluating rectness [Ni24]. In professional settings, AI productivity relies on suggestion ather than time-to-completion, using telemetry models to suppress low-quality ons and minimize cognitive fatigue [Mo24]. However, commercial deployment ires human oversight to meet production standards like accessibility [Mo25].

onmental applications, low-cost IoT sensors face hardware noise and calibration rver-side ML algorithms (e.g., Random Forest, XGBoost, Support Vector on) correct these errors, enabling real-time Air Quality Index (AQI) predictions ng student-built sensors viable for scalable monitoring [Ka23].

isting literature addresses IoT pedagogy, project management, AI coding, and ysis separately, a research gap remains in synthesizing them into a cohesive ducation curriculum. Combining tangible IoT development with advanced AI provides a unified framework to teach web engineering, modern AI coding, and le environmental applications within a globally coordinated initiative [Zi22].

search methodology

nter semester of the 2025/2026 academic year, Bielefeld University of Applied and Arts and WSB Merito University in Gdańsk executed a joint educational Supervised by Prof. Dr. Ing. Grit Behrens and Prof. dr hab. inż. Cezary Orłowski, tive aimed to build practical competencies in distributed data processing systems l-world air quality data.

nental measurements—including NO, CO, NO_2, O_3, SO_2, PM2.5, PM10, and ere sourced via the Airly API (**Fig. 1**). Working with authentic data provided with a production-like operating environment.

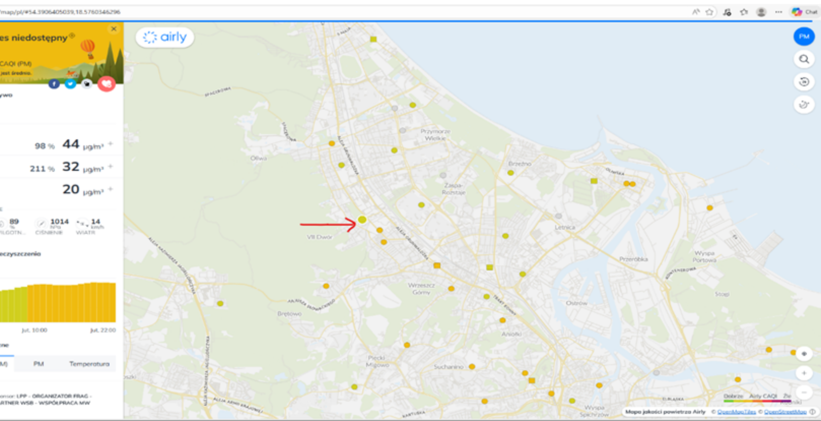


Fig. 1 Airly API: Our Data Source

tudents from WSB Merito University in Gdańsk implemented nine applications, icated to a specific local air quality monitoring station. The course combined al foundations delivered via Moodle with hands-on, experiential learning. g an introduction to project objectives and distributed systems concepts, students he implementation phase using Node-RED for flow-based data pipeline modeling bitMQ for asynchronous message queuing and routing.

led data processing pipeline is presented in Figure 2 (Fig. 2 – Data Flow Steps), ustrates the successive stages of data handling—from data acquisition via the

ough preprocessing and classification, to storage in structures enabling further
This approach allowed students to gain a comprehensive understanding not only
rocessing itself but also of asynchronous communication mechanisms and the
y characteristics of distributed systems.

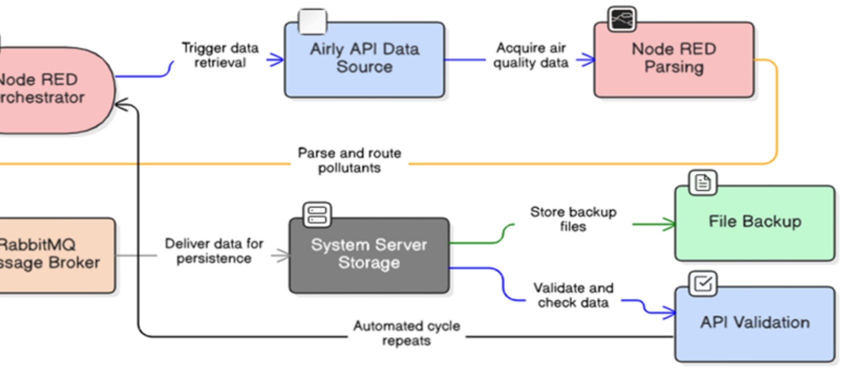


Fig. 2 Data Flow Steps

oject aspect was preparing data for artificial intelligence systems, specifi-cally
-Augmented Generation (RAG) paradigms. Processed RabbitMQ data were
d into structured textual documents to create knowledge bases for Large
e Models (LLMs). Students managed the complete solution lifecycle, including
containerization (docker-compose), Node-RED pipeline develop-ment, system
ng, and technical presentations covering architecture and data flows.

omes were presented in the form of technical presentations covering sys-tem
ure, component configuration, and data flow analysis. Monitoring mechanisms
ntial use cases of the collected data are depicted in Figure 3 (Fig. 3 – Monitoring
g the Data Collected), which highlights both technical and application-oriented
ves.

Fig. 3 – Monitoring and Using the Data Collected

ect yielded nine functional applications and several key pedagogical insights:

oling Synergy: Flow-based tools (Node-RED) combined with message brokers abbitMQ) simplified distributed system design and supported rapid prototyping ibe coding"), particularly for automated container configuration (e.g., docker-mpose).

line Limitations: Fully remote instruction restricted communication and mwork development, indicating that hybrid or in-person formats are better suited collaborative skills.

gineering Standards: While individual projects accelerated implementation, y lacked formal software lifecycle rigor. Future iterations should incorporate ting, version control, and documentation

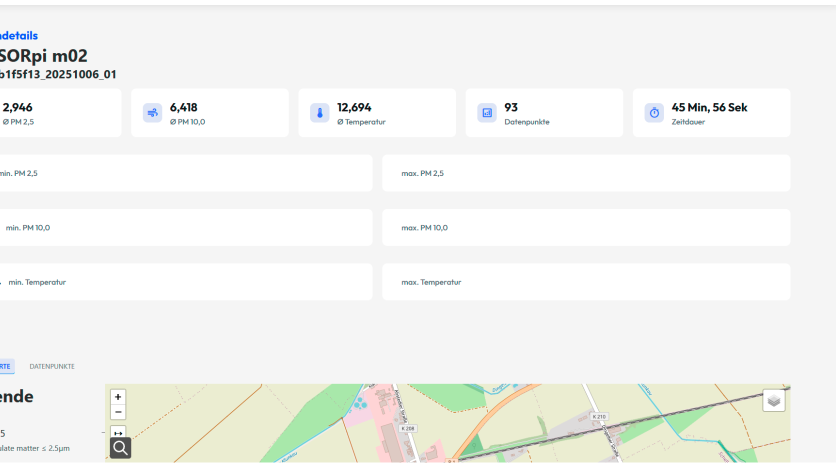
sion, the project represents a valuable example of integrating academic education ctical applications of modern data processing technologies and artificial ce. The incorporation of real-world data sources, industry-relevant tools, and onal collaboration significantly enhanced the quality of the learning process and ed to better preparing students for the challenges of the contemporary labor

egration of Web Engineering and Environmental Educa-tion

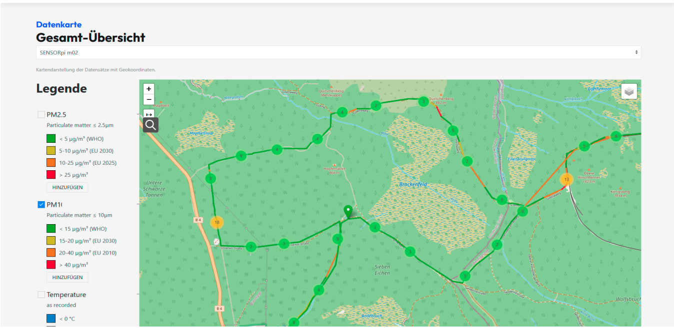
he 2025/2026 winter semester at Hochschule Bielefeld (Campus Minden), the gineering course applied a Project-Based Learning (PBL) approach connecting engineering with healthy air initiatives. Students developed web-based nts for the SmartMonitoring Airquality system, addressing urban air quality data n, visualization, and analysis.

hodology combined theoretical instruction, agile workflows, and functional ng. Following an international exchange lecture by Dr. Łukasz Szałata (Wrocław y of Science and Technology) on air quality metrics (PM2.5, PM10, NO_x, O_3, ure), students translated environmental domain knowledge into technical ents, database models, APIs, and web applications.

rse utilized real-world stationary data from Gdansk (Airly API) alongside y and mobile bicycle-based sensor data from Minden. Incorporating cycling itizen-science dimension, allowing students to analyze spatial-temporal pollution compare routes, and identify high exposure areas. Development was divided into ed strands, including a Researcher Interface featuring map visualizations, route nent, sensor overviews, statistical summaries, and configurable plots for decision-



Web page for route details of mobile bicycle sensor with values for mean PM2.5 of 2.9, 10 of 6.418, mean temperature, number of measurements along the route and duration of the route.



Visualization of a bicycle route in green area around mountain Brocken i Germany in students Webapplication

group developed a Progressive Web Application (PWA) with push notifications ification mechanisms—including achievements, streaks, and leaderboards—to ze regular user participation in environmental data collection.

hievements designed from the student team for motivation for air-quality measurements and used in gamificationed WEB-Application-Prototyp.

software engineering perspective, students applied full-lifecycle practices— requirements analysis, architecture and database modeling, REST APIs, Gitorkflows, issue tracking, and iterative development. The technological stack d the SmartData API, Smart Web Application Components, QL/PostGIS, Payara, Swagger/OpenAPI, and PWA capabilities (Service Push API).

roach integrated three core learning dimensions:

chnical Mastery: Hands-on competencies in web engineering and distributed stems.

main Context: Deeper understanding of urban air quality issues.

vironmental Literacy: Leveraging informatics to transform raw data into visible, ionable insights for sustainability and healthier mobility choices.

commendations

n the evaluation of this joint international initiative, three key sets of ndations are proposed to optimize future educational programs integrating engineering, distributed systems, and environmental analytics:

ftware Engineering Didactics

Balance Prototyping with Rigor: While tools like Node-RED, RabbitMQ, and Docker accelerate architectural understanding through "learning by doing", future iterations must strictly integrate full software lifecycle practices—specifically automated test-ing, systematic documentation, version control, and code quality assurance.

Near-Production Environments: Practical projects should continuously rely on authentic data streams to bridge the gap between academic theory and industrial software deployment.

plication Design in Environmental Engineering

Domain-Centric Design: Environmental context must serve as the structural founda-tion of the system (dictating requirements, data models, and UI) rather than a superfi-cial dataset.

Citizen Science & Engagement: Integrating gamification, push notifications, and mobile sensor data (e.g., bicycle-based measurements) enhances spatial-temporal analysis, encourages user participation, and drives interdisciplinary learning.

lti-University Collaborative Frameworks

Hybrid Instruction Model: Purely remote delivery hinders teamwork and communica-tion. A hybrid approach—combining online collaboration with targeted in-person sessions—is essential for building soft competencies.

Standardized Infrastructure: Joint courses require unified work environments (e.g., shared Git platforms, project management tools), clear role allocation, and aligned evaluation standards to facilitate effective cross-institutional knowledge transfer.

scussion and conclusions

ors clearly emphasize that international cooperation forms the cornerstone of research and educational initiatives. By combining the expertise of academics ents across universities around a shared objective—the development and ntation of the "Air" project—this initiative established an innovative international rk for data processing and forecasting systems.

strength of the project lies in placing the environmental context at the very center ngineering lifecycle: air quality parameters directly shaped the requirements database models, and user interfaces. This domain-driven approach allowed o develop a highly functional tool for data processing and air-quality-aware route tion, while substantially strengthening their practical IT competencies.

e the success and sustainability of future cross-border teaching initiatives, the ighlight several structural insights:

centive Frameworks for Online Collaboration: Managing remote, multi-titutional teams requires significant effort. To boost engagement, academic rricula and examination regulations should be adapted to award extra ECTS dits to participating students and credit additional teaching hours to instructors.

herent Project Structure: Joint university projects demand clearly defined goals, ar role distribution, and intensive student-instructor interaction to overcome the herent challenges of remote delivery.

lti-Format Educational Model: A multi-format teaching approach that tematically reinforces formal software engineering practices across consecutive nesters provides high educational value, preparing students for real-world labor rket challenges.

y, this project demonstrates how merging computer science education with nental awareness creates a valuable, replicable model for international academic tion.

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Green Coding